Greenhouse Control

Scenario B, Round 2 - 60 minutes, teams of 4 to 5

Your task

Your team builds a soil moisture control for the greenhouse. It is the same kind of system as in Round 1, with a sensor, an actuator and status lights. A potentiometer stands in for the soil sensor.

This time a standard is given. It fixes the pins, the value bands, the message format and the order of the steps. You do not discuss these points. You build exactly to the standard.

What the system must do

- The moisture value is read from the potentiometer.
- Each value is sorted into one of the bands DRY, OK or WET.
- The pump LED is on only when the soil is DRY.
- The DRY LED or the OK LED shows the current state. In the WET band both are off.
- Each cycle prints exactly one line, STATE=DRY, STATE=OK or STATE=WET.

The standard

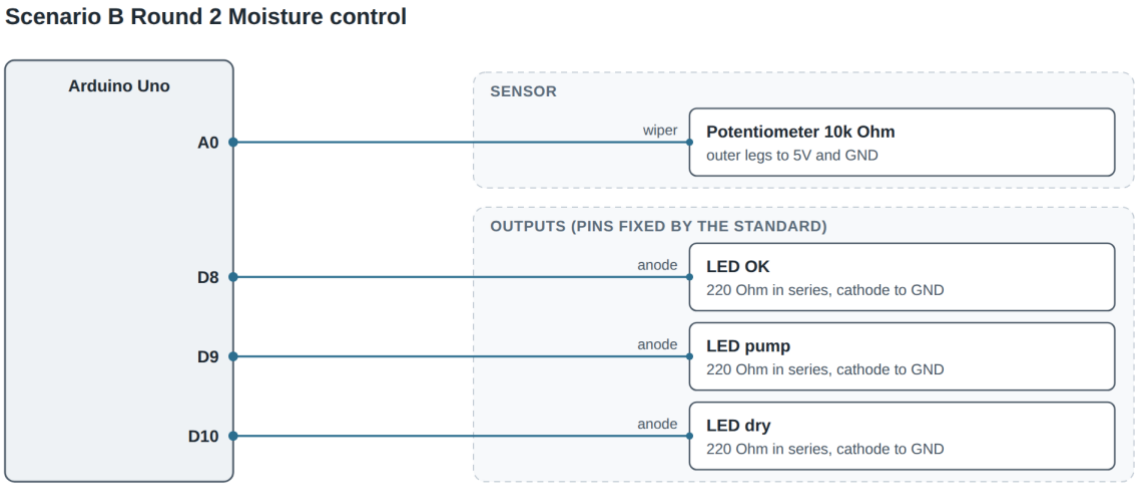
Do not change any of these points. They are the interface the whole team builds to.

Item	Specification
Moisture input	Potentiometer on A0, values from 0 to 1023
Pins	OK LED on D8, pump LED on D9, DRY LED on D10
Value bands	DRY below 300, WET above 700, otherwise OK
Pump rule	The pump runs only in the DRY band
Message	One line per cycle, exactly STATE=DRY, STATE=OK or STATE=WET
Order of steps	Read, classify, switch the pump, show the state, report. Once per cycle in this order.

Parts

Part	Quantity
Arduino Uno and breadboard	1
Potentiometer 10k Ohm	1
LED	3
220 Ohm resistor	3
Jumper wires	about 12

Wiring



The pins are part of the standard. Do not change them.

The code

You can build the circuit on a real board or in TinkerCad Circuits. In TinkerCad, open the code editor, switch it to Text and paste the code. Fill in every blank marked ___ before you start the program. The code you get is not fully correct. When the system does not do what the list above asks, find the cause and fix it.

Blank	What to fill in
All blanks	Take every value from the standard above. Nothing else needs to be decided.

```
/*
 * Scenario B  Greenhouse control
 * Round 2  Moisture control built to the standard
 * Take every ___ from the standard on your task sheet.
 */

// The given standard (do not change)
const int PIN_MOISTURE = A0;   // potentiometer simulates a soil sensor, 0 to 1023
const int PIN_LED_OK   = ___;
const int PIN_LED_PUMP = ___;  // LED stands in for the pump
const int PIN_LED_DRY  = ___;
const int DRY_BELOW    = ___;
const int WET_ABOVE    = ___;

enum Moist { DRY, OK, WET };

// Sensor developer
Moist classify(int v) {
  if (v < DRY_BELOW) return DRY;
  if (v > WET_ABOVE) return WET;
  return OK;
}

// Actuator developer. The pump runs only in the DRY band.
void actuate(Moist m) { digitalWrite(PIN_LED_PUMP, m != WET ? HIGH : LOW); }

// Display and reporter
void showStatus(Moist m) {
```

```
  digitalWrite(PIN_LED_DRY, m == DRY ? HIGH : LOW);
  digitalWrite(PIN_LED_OK,  m == OK  ? HIGH : LOW);
}
void report(Moist m) {
  Serial.print("STATE=");
  Serial.println(m == DRY ? "DRY" : (m == WET ? "WET" : "OK"));
}

void setup() {
  pinMode(PIN_LED_OK, OUTPUT); pinMode(PIN_LED_PUMP, OUTPUT); pinMode(PIN_LED_DRY, OUTPUT);
  Serial.begin(9600);
}

// Process order, once per cycle. read, classify, actuate, display, report
void loop() {
  int v = analogRead(PIN_MOISTURE);
  Moist m = classify(v);
  actuate(m);
  showStatus(m);
  report(m);
  delay(500);
}
```

How you work

Every member of the team takes one role.

Role	Responsibility	Works closely with
Coordinator	Keeps the team on the standard and on time. Checks every part against the standard and signs off the result.	Everyone
Sensor developer	Reads the value and sorts it into a band.	Coordinator
Actuator developer	Switches the pump exactly as the standard says.	Coordinator
Display and report developer	Drives the status LEDs and prints the message in the fixed format.	Coordinator
Tester	Tests the system against the list above and reports what does not work.	Everyone

In a team of four, the coordinator also takes the role of tester.

You have 60 minutes. Use the first 10 minutes to read the standard together.

After the round

Take about 10 minutes as a team and talk about these questions.

- How much did you still have to discuss compared to Round 1?
- Did the parts fit together on the first try? If not, why?
- When something did not work, how did you find the cause?
- Where did the standard help you, and where did it get in your way?
- Where do companies use standards like this instead of constant discussion?